

GAHAN SARAIYA

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PROFESSIONAL SUMMARY

Staff-level Software Engineer with 8+ years of expertise in High-Performance Computing (HPC), parallel programming, and distributed systems architecture. Proven track record in designing and implementing scalable cluster architectures, health telemetry systems, and GPU-accelerated workloads for data center environments. Specialized in System Management Interface (SMI) design, workload orchestration, and performance optimization across heterogeneous computing platforms. Led cross-functional teams in developing enterprise-grade firmware automation frameworks and platform provisioning systems. Strong background in CUDA, ROCm, HIP programming, and cloud-native infrastructure.

TECHNICAL SKILLS

Programming Languages	Python , C, C++, CUDA , HIP , Shell Scripting, SQL, YAML, LaTeX
Parallel & Distributed Computing	CUDA Programming, ROCm, HIP Programming , Multi-GPU Systems, Cluster Architecture Design, Workload Orchestration, Parallel Algorithm Design
Health Telemetry & Monitoring	System Management Interface (SMI), GPU Telemetry, Hardware Metrics, Performance Profiling, In-Band Monitoring, Real-time Health Monitoring
ML/AI Frameworks	TensorFlow, PyTorch, LangGraph, FAISS, TensorRT, PyCUDA, OpenAI/Ver
System Software & Firmware	UEFI, EDK2, Firmware Architecture, Boot Optimization, RAS (Reliability, Availability, Serviceability), CPER, GPU Partitioning
DevOps & Infrastructure	Kubernetes, Docker, Docker Compose, Terraform, Jenkins, GitHub Actions, CI/CD Pipelines, Infrastructure as Code, Platform Provisioning
Cloud & Services	AWS (EC2, S3, Lambda, SageMaker, ECS, EKS), GCP, Azure, Vertex AI, RESTful APIs, JFrog Artifactory, TeamCity, JIRA

EXPERIENCE

Senior System Software Designer at AMD India Pvt Ltd Mar 2024 - Present

- Architected and implemented in-band health telemetry system for GPU clusters using System Management Interface (SMI), enabling real-time monitoring of temperature, power consumption, memory utilization, and compute core metrics across data center deployments.
- Designed and developed dynamic workload orchestration framework for multi-GPU cluster environments, implementing load balancing algorithms that optimize resource allocation based on real-time hardware telemetry and thermal constraints.
- Led performance benchmarking initiatives for HPC devices (GPUs), developing comprehensive test suites covering TensorFlow, PyTorch, and GEMM operations that validate performance against industry standards and identify optimization opportunities.
- Architected parallel programming solutions using ROCm and HIP frameworks, enabling efficient multi-GPU workload distribution and achieving improved cluster utilization through intelligent workload scheduling.
- Designed and implemented RAS (Reliability, Availability, Serviceability) injection mechanisms for MI2XX and MI3XX platforms, establishing health monitoring protocols and automated recovery procedures for production environments.
- Developed cluster health monitoring infrastructure with in-band profiling capabilities, capturing kernel-level performance metrics and system telemetry to enable proactive issue detection and performance optimization.
- Collaborated with cross-functional teams of engineers and researchers to define system architecture requirements, establish best practices for parallel programming, and drive technical decisions for next-generation HPC platforms.

Senior Firmware Development Engineer at Intel Technology India Pvt Ltd Oct 2023 - Mar 2024

- Architected platform provisioning infrastructure for large-scale HPC product automation, designing cluster-based deployment systems that reduced manual configuration time by 70% through automated provisioning workflows.
- Led Platform Orchestration Layer (POL) architecture design for Universal Scalable Firmware (USF), establishing scalable firmware configuration management framework supporting heterogeneous computing platforms.
- Designed and implemented Terraform-based infrastructure automation for Kubernetes cluster environments, enabling rapid deployment of development and testing platforms with infrastructure-as-code practices.
- Integrated Azure OpenAI services into firmware development workflow, implementing automated code review, security analysis, and enhancement suggestions that accelerated development cycles.
- Architected CI/CD pipeline infrastructure using GitHub Actions and CloudBees Jenkins, supporting both bare-metal and containerized deployment targets for continuous integration and testing.
- Optimized firmware boot-time performance for client platforms that reduced boot latency%.

Firmware Development Engineer at Intel Technology India Pvt Ltd Jun 2020 - Sept 2023

- Led architecture design for next-generation client platform at USF Platform Orchestration Layer, owning domain responsibilities for Automation Framework, Platform Provisioning, and Generative AI integration for firmware development.
- Managed end-to-end Software Development Lifecycle (SDLC) including licensing, legal compliance, IP planning, code security, and static code analysis, establishing processes that ensured compliance across all firmware releases.
- Architected and implemented automated security scanning infrastructure using Jenkins, integrating Snyk and Coverity for continuous security analysis and vulnerability detection in firmware codebase.
- Developed automated code quality pipeline using GitHub hooks, implementing Python packaging validation, linting with Bandit, and security analysis that improved code quality metrics by 60%.
- Designed and implemented automated firmware validation framework using TeamCity, enabling continuous functional testing and reducing manual testing overhead by 60%.
- Mentored junior engineers and conducted technical knowledge-sharing sessions on firmware architecture, parallel programming concepts, and system design principles.

Firmware Engineer Intern at Intel Technology India Pvt Ltd Jun 2019 - May 2020

- Developed BIOS firmware for client products using EDK2 and UEFI frameworks, contributing to debugging solutions and OpenBoard Package integration.

Teaching Assistantship at Nirma University Aug 2018 - May 2019

- Data Mining and Visualization

Software Developer at Quixom Technology Jul 2017 - Apr 2018

- SaaS-Employee management system (chat, surveys, benchmark, analytics...) and Restful API Development
- Socket programming in python to receive continuous data from IoT device and display chunks of data meaningfully
- Streaming service with Web-Portal, DRF and Kodi across platforms

Freelance Full-stack Developer 2015 - Apr 2019

- Cloud solution for Data gathering and webscraping with selenium and GCP.
- Designing template and banners for High resolution prints

- Inventory management system for Book store with Django.

PROJECTS

Agentic AI for Software Engineering Modernization

2025 - Present

Description: Modernized enterprise repository workflows, code reviews, and project management by engineering multi-agent GenAI systems.

- Developed tool-compatible custom skills, automated Git hooks, and multi-modal agents leveraging Claude, Cursor, and OpenAI API frameworks.
- Engineered rigorous security compliance, sandboxed execution verification, and real-time LLM token spend/quota restriction mechanisms.

Context-Aware Task Routing & Triage with LangGraph

2025 - Present

Description: Led agile sprint cycles and managed timeline deliveries to develop an end-to-end, context-aware ticket/bug classification and routing engine.

- Built graph-based LLM logic using LangGraph for cyclic agent routing, integrated with semantic search using FAISS vector databases and specialized embeddings.
- Integrated diverse data sources (GitHub commits, Jira tickets, documentation history) with multiple APIs including OpenAI, Vertex AI, and sandboxed local LLMs.
- Formulated a persona-based LLM counselor model that adapts summaries and technical advice, operating behind a strict quality matrix verification gate.

DevOps Management for Cluster Validation

2025 - Present

Description: Established continuous integration infrastructure for automated validation of large-scale heterogeneous HPC and AI computing clusters.

- Automated secure telemetry log shipping to S3 buckets, integrating custom pipeline controls and IAM roles for granular resource tracking.
- Structured fine-grained security policy and compliance enforcement across cloud networks and multi-region regulatory domains.

GPU Cluster Performance Analysis & Health Telemetry System

2024 - Present

Description: Enterprise-grade solution for in-band profiling and health monitoring of GPU clusters in data center environments

- **Cluster Health Telemetry Architecture:** Architected and implemented comprehensive in-band profiling system using System Management Interface (SMI) for real-time GPU cluster health monitoring, capturing telemetry data across temperature, power, memory, and compute utilization metrics.
- **Dynamic Cluster Workload Orchestration:** Designed intelligent workload scheduling system that dynamically adapts to real-time cluster resource availability, implementing load balancing algorithms that optimize multi-GPU workload distribution and improve overall cluster efficiency.
- **Distributed Health Monitoring Infrastructure:** Engineered scalable telemetry collection framework supporting hundreds of GPU nodes, implementing in-band monitoring channels that minimize performance overhead while providing comprehensive hardware metric visibility.
- **Performance Profiling & Analysis:** Developed advanced profiling infrastructure capturing kernel-level performance metrics, call stack timings, and execution patterns to identify bottlenecks and optimize parallel program performance across cluster deployments.
- **Unified Analytics Platform:** Innovated methodology to aggregate and correlate multi-dimensional metrics (thermal, power, utilization, kernel timings) into unified health scores, enabling predictive maintenance and proactive cluster management.

Platform Orchestration & Cluster Provisioning System

2023 - 2024

Description: Automated platform provisioning infrastructure for large-scale HPC cluster deployment

- Architected cluster provisioning framework supporting remote BMC and BIOS firmware flashing across distributed server nodes, enabling zero-touch deployment for pre-production reference boards.
- Designed automated OS deployment pipeline integrated with Jenkins, supporting artifact management through JFrog Artifactory and RedFish API-based system configuration.
- Implemented infrastructure-as-code practices using Terraform for cluster environment provisioning, reducing deployment time from days to hours.

Universal Scalable Firmware (USF) - Platform Orchestration Layer 2022 - 2024

Description: Architecture design of scalable firmware configuration management for heterogeneous computing platforms

- Led architecture design of Platform Orchestration Layer enabling unified firmware configuration across diverse hardware platforms, contributing to UEFI industry standard development.
- Reference: [Firmware Configuration – Past, Present, and Future](#) (UEFI DevCon 2023)

Firmware Automation Framework 2020 - Present

Description: Enterprise automation framework for large-scale firmware development and validation

- Architected end-to-end automation framework supporting thousands of silicon devices, implementing zero-touch BIOS configuration and automated validation workflows.
- Established SDLC processes including security scanning, static code analysis, legal compliance, and licensing management integrated into CI/CD pipelines.
- Designed continuous deployment infrastructure using Jenkins and GitHub Actions, managing release artifacts through JFrog Artifactory with automated testing and validation.
- Open Source: <https://github.com/intel/xml-cli>

EDUCATION

M.Tech Computer Science Nirma University, GPA: 8.49	Jul 2018 - Jun 2020
B.E. in Computer Science and Engineering CGPA: 8.08	Aug 2013 - Apr 2017 Gujarat Technological University,

CERTIFICATIONS

2025	AMD ROCm Star - Application Developer by AMD Adaptive Computing Training Team
2025	On-Device AI by Qualcomm & DeepLearning.ai
2025	Essential Terraform in AWS
2021	Product Assurance and Security Yellow Belt - Software
2020	Natural Language Processing with Classification and Vector Spaces Coursera S9DGGMCDMBGQ
2020	Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization Coursera SMWQJKJYJW6G
2020	Structuring Machine Learning Projects - Coursera SEJML7H3TN5Z
2020	Neural Networks and Deep Learning - Coursera S2TETZHTLUSL
2020	Machine Learning - Coursera XVNXNPPZ3JMA

ACHIEVEMENTS

2026	Top 10 in AI Hackathon: Hack2Skill Prompt Wars by Google (In-person, Bangalore)
2026	Rank #3 in Virtual Prompt Wars Hackathon
2023	Firmware Configuration Presentation at UEFI Developers Conference & Plugfest
2023	Tech Talk on Harnessing Multiprogramming across CPU, GPU and QPU (Quantum Processing Unit)
2021	Expert Lecture on CUDA Programming
2019-Present	4000+ reputation on Stackoverflow
2017-Present	Active volunteer on Peer Learning, Cultural and team building activity
2018	Rank #1 for Python challenges on hackerrank
2018	Rank #112 from 2 lacs+ participants in Techgig's National Coding Contest 2018

PUBLICATIONS

• HIP Parallel Programming with ROCm Lecture Series, YouTube	2025
• Beyond the Power — Data Center & BMC Lecture Series, YouTube	2025
• System Design for Parallel Programming Lecture Series, YouTube	2025
• Firmware Configuration – Past, Present, and Future . Presentation, UEFI DevCon 2023	2023